Exercise Sheet

Lecture 4

Matrices of Linear Maps; Change of Basis

Throughout, bases are ordered and $\mathcal{M}(T)$ denotes the matrix of T with respect to the bases in force. Tags indicate the flavour of each problem; a $\star$ marks a harder one.

Core tutorial route

Work **4.1, 4.7, 4.11, 4.14, 4.15, 4.16, 4.17, and 4.20** in that order (about 60–75 minutes before discussion). This eight-problem route checks matrix construction, the composition argument, invertibility, map-space dimension, finite isomorphism structure, a full similarity calculation, coordinate reconstruction, and an invariant proof.

Additional practice: 4.3–4.4, 4.8–4.10, 4.12–4.13, 4.18–4.19. **Bridge:** 4.2, 4.5, 4.6 (polynomial calculus). **Extension:** 4.21–4.23.

Matrices of maps

Exercise 4.1 (computational) Let $T \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2)$ be $T(x, y, z) = (x + 2y - z,\ 3x + z)$. Write $\mathcal{M}(T)$ with respect to the standard bases, and use it to compute $[T(1, 1, 1)]$.

Exercise 4.2 (computational) Let $D \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be differentiation. Compute $\mathcal{M}(D)$ and $\mathcal{M}(D)^2$ in the standard basis $1, t, t^2$, and identify the operator that $\mathcal{M}(D)^2$ represents.

Exercise 4.3 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2)$ be $T(x, y) = (2x - y,\ x + 3y)$. Write $\mathcal{M}(T)$ in the standard basis and use it to compute $[T(1, 2)]$.

Exercise 4.4 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ be $T(x, y) = (x - y,\ 2x + y,\ 4y)$. Write $\mathcal{M}(T)$ with respect to the standard bases and compute $[T(3, -1)]$.

Exercise 4.5 (computational) Let $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be $(Tp)(t) = p(t) + t\,p'(t)$. Compute $\mathcal{M}(T)$ in the standard basis $1, t, t^2$ and use it to evaluate $T(2 - t + 3t^2)$.

Exercise 4.6 (computational) Let $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}))$ be differentiation, with $\mathcal{M}(D) = \begin{pmatrix} 0&1&0&0\\ 0&0&2&0\\ 0&0&0&3\\ 0&0&0&0 \end{pmatrix}$ in the standard basis $1, t, t^2, t^3$. Compute $\mathcal{M}(D)^3$ and find the smallest k with $D^k = 0$.

Exercise 4.7 (proof) Suppose $S, T \in \mathcal{L}(V)$ where $\dim V = n$, with matrices $\mathcal{M}(S), \mathcal{M}(T)$ in a fixed basis. First derive $[STv] = \mathcal{M}(S)\mathcal{M}(T)[v]$ for an arbitrary v and explain why this forces $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$. Then prove that $ST = TS$ as operators if and only if $\mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(T)\mathcal{M}(S)$ as matrices.

Exercise 4.8 (proof) Fix bases of V and W. Prove that for $S, T \in \mathcal{L}(V, W)$ and $a \in \mathbb{F}$, $\mathcal{M}(aS + T) = a\,\mathcal{M}(S) + \mathcal{M}(T)$.

Invertibility and isomorphism

Exercise 4.9 (computational) Decide whether $T(x, y) = (2x + y,\ 4x + 2y)$ on $\mathbb{R}^2$ is invertible. If so give T^{-1}; if not give a nonzero vector in null T.

Exercise 4.10 (computational) Show that $1,\ 1-t,\ (1-t)^2$ is a basis of $\mathcal{P}_2(\mathbb{R})$ by transporting the question to $\mathbb{R}^3$.

Exercise 4.11 (computational) Decide whether $T(x,y) = (3x+5y,\ x+2y)$ on $\mathbb{R}^2$ is invertible; if so, give T^{-1} explicitly.

Exercise 4.12 (computational) Decide whether $T(x,y,z) = (x+2y+3z,\ 2x+4y+6z,\ x+y+z)$ on $\mathbb{R}^3$ is invertible. If not, give a nonzero vector in $\operatorname{null} T$.

Exercise 4.13 (computational) Show that $1+t^2,\ t+t^3,\ 1+t,\ t^2$ is a basis of $\mathcal{P}_3(\mathbb{R})$ by transporting the question to $\mathbb{R}^4$.

Exercise 4.14 (proof) Prove that if V and W are finite-dimensional, then $\dim \mathcal{L}(V,W) = (\dim V)(\dim W)$.

Exercise 4.15 (proof) Let V be finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent: (i) T is injective; (ii) T is surjective; (iii) T is invertible.

Change of basis and similarity

Exercise 4.16 (computational) Let $T(x,y) = (x,\ -y)$ on $\mathbb{R}^2$ (reflection in the x-axis), with standard matrix $A = \operatorname{diag}(1,-1)$. Find its matrix in the basis $\mathcal{B}' = \big((1,1),(1,-1)\big)$ using $A' = P^{-1}AP$.

Exercise 4.17 (computational) In $\mathbb{R}^2$ let $\mathcal{B}$ be the standard basis and $\mathcal{B}' = \big((2,1),(1,1)\big)$. A vector has new coordinates $[v]_{\mathcal{B}'} = (1,2)$; find $[v]_{\mathcal{B}}$ (i.e. v itself). Also find the $\mathcal{B}'$-coordinates of $w = (3,0)$.

Exercise 4.18 (computational) Let $T \in \mathcal{L}(\mathbb{R}^2)$ have standard matrix $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$. Find its matrix in the basis $\mathcal{B}' = \big((1,0),(1,1)\big)$ using $A' = P^{-1}AP$.

Exercise 4.19 (computational) Using trace and determinant only, decide whether each pair of matrices can possibly be similar: (a) $\left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right)$; (b) $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 2 & 0 \\ 1 & 2 \end{smallmatrix}\right)$.

Exercise 4.20 (proof) Prove that similar matrices have the same determinant and the same trace.

Exercise 4.21 ($\star$) Suppose A is similar to B and B is similar to C. Prove A is similar to C, and deduce that being similar is an equivalence relation on $n \times n$ matrices.

Exercise 4.22 ($\star$) Suppose A and B are $n \times n$ matrices with A invertible. Prove that AB and BA are similar, and conclude $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Exercise 4.23 ($\star$) Give two 2×2 matrices with the same trace and the same determinant that are nevertheless *not* similar, and prove they are not similar.